

Sprint 1

Why ISNs are random, not 0.

There's less of a chance of multiple nodes having the same ISN.

What happens if the SYN|ACK is lost.

The server retransmits.

Why `recv()` on a UDP socket returns 0 bytes in some cases but means something different from TCP.

In TCP, a 0 means the connection has been closed. In UDP, it could mean the datagram received was 0 bytes long.

Why the server uses `recvfrom()` instead of `recv()`.

UDP does not maintain a connection like TCP, so it needs to be told where its receiving from and sending to.

Demonstration

<pre>Starting server... port = 8008 log = logs/server_2026-04-06_14-31-58.log Testing address 127.0.0.1. Socket setup for 127.0.0.1:8008. recvfrom 127.0.0.1. 21 / 65535 Fin 21 / 21 Recieved SYN. Sending ACK... 21 / 21 Sent 21 bytes. recvfrom 127.0.0.1. 21 / 65535 Fin 21 / 21 Handshake complete.</pre>	<pre>Starting client... port = 8008 IP = 127.0.0.1 log = logs/client_2026-04-06_14-32-01.log file = Socket setup for 127.0.0.1. Sending SYN... 21 / 21 Sent 21 bytes. recvfrom 127.0.0.1. 21 / 65535 Fin 21 / 21 Recieved ACK. Sending ACK... 21 / 21 Sent 21 bytes. Handshake complete. Exiting... moit@moitlegion:~/school/CSC4200-Networks/Github C</pre>
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Sprint 2 - 1

Why sequence numbers are byte-based.

Because packets may be different lengths, seen below.

What a duplicate ACK means and when the server sends one.

A duplicate ACK tells the client that its sent the packets in the wrong order.

What happens if the client retries more than `MAX\_RETRIES` times.

It's hard-coded not to, but all it would do is waste time on a possible dead connection.

Why you must loop on `recvfrom()` — what if you get an out-of-order packet while waiting for a specific ACK?

We may accidentally get the next packet before the ACK, so we keep listening for the ACK.

Client Log

```
2026-04-05-14-19-22 SEND SEQ=563831546 ACK=0 0 1 0 LEN=0
2026-04-05-14-19-22 RECV SEQ=2126880684 ACK=563831547 1 1 0 LEN=0
2026-04-05-14-19-22 SEND SEQ=2126880684 ACK=563831547 1 0 0 LEN=0
2026-04-05-14-19-22 SEND SEQ=563831547 ACK=0 0 0 0 LEN=53248
2026-04-05-14-19-22 RECV SEQ=0 ACK=563884795 1 0 0 LEN=0
2026-04-05-14-19-22 SEND SEQ=563884795 ACK=0 0 0 0 LEN=53248
2026-04-05-14-19-22 RECV SEQ=0 ACK=563938043 1 0 0 LEN=0
2026-04-05-14-19-22 SEND SEQ=563938043 ACK=0 0 0 0 LEN=53248
2026-04-05-14-19-22 RECV SEQ=0 ACK=563991291 1 0 0 LEN=0
2026-04-05-14-19-22 SEND SEQ=563991291 ACK=0 0 0 0 LEN=53248
2026-04-05-14-19-22 RECV SEQ=0 ACK=564044539 1 0 0 LEN=0
2026-04-05-14-19-22 SEND SEQ=564044539 ACK=0 0 0 0 LEN=53248
2026-04-05-14-19-22 RECV SEQ=0 ACK=564097787 1 0 0 LEN=0
2026-04-05-14-19-22 SEND SEQ=564097787 ACK=0 0 0 0 LEN=49002
2026-04-05-14-19-22 RECV SEQ=0 ACK=564146789 1 0 0 LEN=0
2026-04-05-14-19-22 SEND SEQ=564146789 ACK=0 0 1 LEN=53248
2026-04-05-14-19-22 RECV SEQ=0 ACK=564200037 1 0 0 LEN=0
```

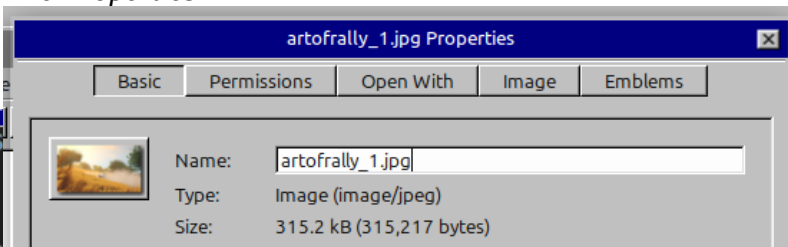
Server Log

```
2026-04-05-14-19-22 RECV SEQ=563831546 ACK=0 0 1 0 LEN=0
2026-04-05-14-19-22 SEND SEQ=2126880684 ACK=563831547 1 1 0 LEN=0
2026-04-05-14-19-22 RECV SEQ=2126880684 ACK=563831547 1 0 0 LEN=0
2026-04-05-14-19-22 RECV SEQ=563831547 ACK=0 0 0 0 LEN=53248
2026-04-05-14-19-22 SEND SEQ=0 ACK=563884795 1 0 0 LEN=0
2026-04-05-14-19-22 RECV SEQ=563884795 ACK=0 0 0 0 LEN=53248
2026-04-05-14-19-22 SEND SEQ=0 ACK=563938043 1 0 0 LEN=0
2026-04-05-14-19-22 RECV SEQ=563938043 ACK=0 0 0 0 LEN=53248
2026-04-05-14-19-22 SEND SEQ=0 ACK=563991291 1 0 0 LEN=0
2026-04-05-14-19-22 RECV SEQ=563991291 ACK=0 0 0 0 LEN=53248
2026-04-05-14-19-22 SEND SEQ=0 ACK=564044539 1 0 0 LEN=0
2026-04-05-14-19-22 RECV SEQ=564044539 ACK=0 0 0 0 LEN=53248
2026-04-05-14-19-22 SEND SEQ=0 ACK=564097787 1 0 0 LEN=0
2026-04-05-14-19-22 RECV SEQ=564097787 ACK=0 0 0 0 LEN=49002
2026-04-05-14-19-22 SEND SEQ=0 ACK=564146789 1 0 0 LEN=0
2026-04-05-14-19-22 RECV SEQ=564146789 ACK=0 0 1 LEN=53248
2026-04-05-14-19-22 SEND SEQ=0 ACK=564200037 1 0 0 LEN=0
```

Checksum Verification

```
moit@moitlegion:~/school/CSC4200-Networks/Github Classroom/CS
s/artofrally_1.jpg && md5sum downloads/artofrally_1.jpg
1d5eb4cb05a5e8933d0d20b9fb24ce84 res/artofrally_1.jpg
1d5eb4cb05a5e8933d0d20b9fb24ce84 downloads/artofrally_1.jpg
moit@moitlegion:~/school/CSC4200-Networks/Github Classroom/CS
```

File Properties



Server Console

```

moit@moitlegion: ~/school/CSC4200-N
Sending ACK...
  16 / 16
  Sent 16 bytes.
recvfrom 127.0.0.1.
  16 / 212992
  New size: 16.
Fin 16 / 16
Handshake complete.
Waiting for data...
recvfrom 127.0.0.1.
  53264 / 212992
  New size: 53264.
Fin 53264 / 53264
Receiving artofrally_1.jpg to downloads/artofrally_1.jpg
Sending ACK 0.
  16 / 16
  Sent 16 bytes.
recvfrom 127.0.0.1.
  53264 / 212992
  New size: 53264.
Fin 53264 / 53264
Sending ACK 1.
  16 / 16
  Sent 16 bytes.
recvfrom 127.0.0.1.
  53264 / 212992
  New size: 53264.
Fin 53264 / 53264
Sending ACK 2.
  16 / 16
  Sent 16 bytes.
recvfrom 127.0.0.1.
  53264 / 212992
  New size: 53264.
Fin 53264 / 53264
Sending ACK 3.
  16 / 16
  Sent 16 bytes.
recvfrom 127.0.0.1.
  53264 / 212992
  New size: 53264.
Fin 53264 / 53264
Sending ACK 4.
  16 / 16
  Sent 16 bytes.
recvfrom 127.0.0.1.
  49018 / 212992
  New size: 49018.
Fin 49018 / 49018
Sending ACK 5.
  16 / 16
  Sent 16 bytes.
recvfrom 127.0.0.1.
  53264 / 212992
  New size: 53264.
Fin 53264 / 53264
Sending ACK 6.
  16 / 16
  Sent 16 bytes.
All packets received.

```

Client Console

```

moit@moitlegion: ~/school/CSC4200-N
Reading 315217 bytes from res/artofrally_1.jpg.
Read 315217 of 315217 bytes.
File too big. Splitting into 6 packets...
Sending packet 0 with offset 0 to 53247 (53248).
  53264 / 53264
  Sent 53264 bytes.
Getting ACK 0...
recvfrom 127.0.0.1.
  16 / 212992
  New size: 16.
Fin 16 / 16
Sending packet 1 with offset 53248 to 106495 (53248).
  53264 / 53264
  Sent 53264 bytes.
Getting ACK 1...
recvfrom 127.0.0.1.
  16 / 212992
  New size: 16.
Fin 16 / 16
Sending packet 2 with offset 106496 to 159743 (53248).
  53264 / 53264
  Sent 53264 bytes.
Getting ACK 2...
recvfrom 127.0.0.1.
  16 / 212992
  New size: 16.
Fin 16 / 16
Sending packet 3 with offset 159744 to 212991 (53248).
  53264 / 53264
  Sent 53264 bytes.
Getting ACK 3...
recvfrom 127.0.0.1.
  16 / 212992
  New size: 16.
Fin 16 / 16
Sending packet 4 with offset 212992 to 266239 (53248).
  53264 / 53264
  Sent 53264 bytes.
Getting ACK 4...
recvfrom 127.0.0.1.
  16 / 212992
  New size: 16.
Fin 16 / 16
Sending packet 5 with offset 266240 to 315241 (49002).
  49018 / 49018
  Sent 49018 bytes.
Getting ACK 5...
recvfrom 127.0.0.1.
  16 / 212992
  New size: 16.
Fin 16 / 16
Sending packet 6 with offset 319488 to 372735 (53248).
  53264 / 53264
  Sent 53264 bytes.
Getting ACK 6...
recvfrom 127.0.0.1.
  16 / 212992
  New size: 16.
Fin 16 / 16
Exiting...
moit@moitlegion:~/school/CSC4200-Networks/Github Classroom

```

Interrupt Test

<pre> Sent 16 bytes. recvfrom 127.0.0.1. 53264 / 212992 New size: 53264. Fin 53264 / 53264 Sending ACK 1. 16 / 16 Sent 16 bytes. recvfrom 127.0.0.1. 53264 / 212992 New size: 53264. Fin 53264 / 53264 *****Test Interrupt***** recvfrom err: Resource temporarily unavailable Received 4294967295 bytes. recvfrom 127.0.0.1. 53264 / 212992 New size: 53264. Fin 53264 / 53264 Sending ACK 2. 16 / 16 Sent 16 bytes. recvfrom 127.0.0.1. 53264 / 212992 </pre>	<pre> New size: 16. Fin 16 / 16 Sending packet 1 with offset 53248 to 106495 (53248). 53264 / 53264 Sent 53264 bytes. Getting ACK 1... recvfrom 127.0.0.1. 16 / 212992 New size: 16. Fin 16 / 16 Sending packet 2 with offset 106496 to 159743 (53248). 53264 / 53264 Sent 53264 bytes. Getting ACK 2... recvfrom err: Resource temporarily unavailable Received 4294967295 bytes. Sending packet 2 with offset 106496 to 159743 (53248). 53264 / 53264 Sent 53264 bytes. Getting ACK 2... recvfrom 127.0.0.1. 16 / 212992 New size: 16. Fin 16 / 16 </pre>
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